Alexandra Neagu

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Education

PhD researching Large Language Models for STEM Education, Imperial College London ***2024 – 2028***

- Home student, supervised by Dr Peter B Johnson & Dr Rhodri Nelson
- **Research Question:** *"To what extent can we utilise existing GenAI tools to provide effective dialogic feedback, whilst aligning with educational values and ensuring domain-specific technical accuracy?"*
- Full-stack developer on Lambda Feedback, an online self-study platform currently used in Imperial College, with a focus on integrating AI for conversational feedback on student real-time progress
- Graduate Teaching Assistant for Machine Learning for Data Science, Machine Reasoning, and Robotics Modules
- **Publications:**

Neagu, A., Messer, M., Johnson, P.B. and Nelson, R., 2026, June. "How Do I ...?": Procedural Questions Predominate Student-LLM Chatbot Conversations. In International Conference on Artificial Intelligence in Education 2026. (under review)

Sölch, M., Neagu, A., Messer, M., et al., 2026, June. µEd API: Towards A Shared API for EdTech. In ACM Conference on Learning@ Scale 2026. (under review)

Neagu, A., Johnson, P.B. and Nelson, R., 2025, July. Chatbots for Dialogic Feedback during Self-Study: The Importance of Contextual Information. In EERN 2025 Symposium.

MEng Electronic and Information Engineering, Imperial College London ***2020 – 2024***

- Relevant Year Modules:

-Self-Organised Multi-Agent Systems & Artificial Intelligence *(Inter-agents and intra-environment communications, Game Theory, Graph Theory and Knowledge Representation)*	-Machine Learning *(Learning Algorithms, Neuroscience)* -Privacy Engineering*(Anonymisation, Differential Privacy)* -Computer Vision and Robotics *(SLAM)*

- Home student; First Class Honours overall
- Teaching Assistant for Machine Reasoning Module
- Awarded the *Imperial College President's Undergraduate Scholarship* for academic excellence in Sixth Form
- Final Year Project: *"Enhancing Puzzle-Solving with User-Tailored AI Assistance: A Study on Nonogram Learning Outcomes"* [supervised by Dr Nicole Salomons] - explored how LLMs can be used to provide tailored hints and feedback based on the user's skills at solving an unknown puzzle game developed in Unity.

International Baccalaureate Diploma, Sevenoaks School, Kent ***2018 – 2020***

- Achieved *39/45 marks;* Was Coordinator of students' Coding Club & worked on building kits for learning coding.
- *Subjects HL:* Mathematics (7), Physics (7), Geography (6); *SL:* English Literature (5), German B (5), Mandarin B (6)

Languages, Technical Skills & Certifications

Languages: English (fluent), Romanian (native), German (limited working proficiency, B1-B2), Mandarin (limited working proficiency, HSK 4 - 5), French (elementary proficiency, A1- A2)

Skills: Python, TypeScript, Docker, SQL, GraphQL, Pytorch, C++, C#, Unity, Infrastructure as Code (Pulumi), Git, Bash

AWS & Cloud Certifications achieved: AWS Certified Machine Learning Speciality, AWS Solutions Architect Associate, AWS Certified Cloud Practitioner, HashiCorp Certified: Terraform Associate (003)

Industrial Experience

Professional Services Intern, Amazon Web Services ***Apr-Sept 2023 & Jul-Sept 2022***

6 months industrial placement; Developed skills in cloud services and infrastructure as code.

- Presented and debated my solutions, documented my code and development decisions for the senior consultants and developers; Worked in a global team across EMEA and the US.

Project 1: Set up a Monitoring and Alarming System of global EC2 Instance Fleet for the Managed Environments for Outside Work, a fully managed virtual desktop service; Used AWS services to analyse and visualise metrics. Deployed and provisioned the system resources through AWS CDK, an Infrastructure-as-Code tool.

- *Main challenges: addressing the scalability of the data analysis to the global scale of the Instance Fleet, optimising the infrastructure for best suited data preparation and ingestion approaches.*

Project 2: Built a Serverless Infrastructure for a Text-to-Speech conversion and generation system of natural voices using AWS Polly; Set up a web-app using React Typescript and AWS Amplify.

- *Main challenges: facilitating the infrastructure with simple and secure user interface for interactions with the voice generating pipeline, Deploying the AWS cloud infrastructure as code using Terraform.*

3 months summer internship; Learned about AWS Best Practices in Cloud Security & Infrastructure.

Project: Transferred a Customer Cloud-Ready Assessment onto an online AWS Assessment Tool and created my perspective of the web assessment interface with AWS Amplify, DynamoDB, React and GraphQL.

- *Main challenges: comprehending the compliance guidance and best practices of Landing Zones on AWS.*

Technical Projects and Achievements (for more details on each project, please check out my personal page)

Teaching Simulated Learner Politeness in Language Learning **[LearnLab Carnegie Mellon]** ***Aug 2025***

- A 1-week intensive course focused on creating technology-enhanced learning experiments and building intelligent tutoring systems (ITS) at Carnegie Mellon University, Pittsburgh, USA
- Trained a simulated student using the Knowledge-Learning-Instruction (KLI) framework to learn politeness as knowledge components by interacting with an ITS tutor in a Chinese-learning exercise

Avatour* – VR mediated Robot Telepresence** **[Human-Centred Robotics 4th Year Module]** ***Jan-Mar 2024

- Research oriented team project: engineered a VR telepresence system with a 360° camera on a Boston Dynamics SPOT robot for live remote exploration, whether due to health constraints or financial limitations
- Discovered user prior experience highly influenced their preference between traditional 2D and VR interfaces, contesting our initial hypothesis that VR would enhance trust and immersion

CharIOT* - Cleanroom Monitoring System** **[Embedded Systems 3rd Year Module]** ***Jan-Mar 2023

- An embedded system containing IoT devices based on Raspberry Pi. The goal is to monitor the quality and reliability of a research space or cleanroom for microchip design (temperature, pressure, humidity, particle pollution).
- Built a web app on AWS Amplify that displays the sensor readings from the IoTs and triggers web and physical alarms for out-of-bound readings. Users can define parameter bounds that get synchronised in all IoTs' settings over TCP.

ZeroToOne* - AR/VR Circuit Design Simulator** **[ICL ICHack Hackathon]** ***Feb 2022

- An MVP of an AR/VR circuit studio deployed on an android application, developed in Unity with Vuforia and C#. Used a Phone compatible headset for VR and the phone screen for AR user interfaces. The application was presenting a step-by-step guide for building an example circuit.
- This project was aimed for professors to give out remote guidance to students during their hardware laboratories.

External Courses ***2020 – 2025***

- Undergoing 'Foundations of Engineering Education' course online at State University of New York Buffalo. *[2025]*
- Completed a 2-day Nvidia course on Accelerating CUDA C++ applications with multiple GPUs. *[2024]*
- Completed a 3-day Computer Vision Course from Machine Learning University on AWS SageMaker. *[2023]*
- Completed the 30 hours AWS Certified SysOps Administrator Associate Course on Udemy. *[2023]*
- Completed 3 courses part of the Internet of Things and Embedded Systems Specialization for Raspberry Pi from University of California, Irvine on Coursera. *[2021]*
- Completed a 4-week course in XR (VR/AR) development in Unity from University of Michigan, on Coursera in which I had to adapt a self-made 3D scene in both VR and AR using specific libraries for Unity. *[2020]*

Extracurricular Interests & Volunteering

Participating in Hackathons	**Lazy Browsing [ICL ICHack Hackathon]** – a proof-of-concept browser extension using *browser-use* that allows AI agents to autonomously search over various sites for specific shopping items with your given requirements. [Python, JS]	***Feb 2025***
	Fan Mail Digitalization [AWS Interns Hackathon] – a serverless infrastructure on AWS for digitalizing, translating, filtering, and displaying fan letters during concerts. [AWS Lambda-Textract-Translate-Comprehend-Polly, Python]	***Aug 2023***
	Solar Car Educational Game [WSET Hack-her-thon] – best game award, only-girls team, to teach children the importance of solar energy development through a web-game with a hardware joystick and solar panel. [HTML, Arduino]	***Mar 2021***
Vice President Industry EESoc, Imperial College London	▪ Built collaborations between companies and the Electrical Engineering Society (EESoc); Organised industry talks, webinars, and lead the organisation of one of the largest student-led Careers' Fair in Imperial. ▪ Developed event management skills and improved organisational skills.	***2021-2023***
Teacher Volunteer, Sevenoaks School	▪ Gathered funds through 7 months of self-organised events in Sevenoaks to support the students at SKSN School, Rajasthan, India. ▪ Prepared teaching materials and interactively taught Geography and English.	***Sept 2018 – Apr 2019***
Travelling	▪ I enjoy encountering many different cultures; Travelled around Europe, USA, China, Peru, India, Japan, Madagascar, Costa Rica, Tanzania, Egypt…	***throughout***